RAJALAKSHMI ENGINEERING COLLEGE

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C2

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|---|---|
| **Started on** | Thursday, 17 September 2026, 8:26 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 8:26 AM |
| **Time taken** | 30 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

#define MAX_N 200005

int counts[MAX_N] = {0};

int main() {
    int n;

    // Read the total number of elements
    if (scanf("%d", &n) != 1) return 0;

    int current_val;
    int duplicate = -1;

    // Read the elements from the following input lines
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &current_val) != 1) return 0;

        counts[current_val]++;

        // If the element has been seen before, it is our duplicate
        if (counts[current_val] > 1) {
            duplicate = current_val;
        }
    }

    // Output the duplicate element
    if (duplicate != -1) {
        printf("%d\n", duplicate);
    }

    return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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